

Discrete Structures

Lecture # 3

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Recap

Negation, AND, OR, Exclusive Or with Examples

Logical Equivalence with Examples

De-Morgan's Law with Examples

Tautology with Examples

Contradiction with Examples

Laws of Logic (Homework)!

Today's Topics

Conditional Statement (Implication)

Examples with Exercise

Implication equivalence

Converse, Inverse, and Contrapositive

Bi-Conditional Statement

Conditional Statements

- “If you earn an **A** in Math, then I’ll buy a computer.”
- If **p** and **q** are statement variables, the conditional of **q** by **p** is “If **p** then **q**” or “**p** implies **q**” and is denoted **p**  **q**.

Conditional Statement or Implication

- The arrow “” is the conditional operator

p is called the **hypothesis** (or **antecedent**)

q is called the **conclusion** (or **consequent**)

Truth Table for Implication

$$p \rightarrow q$$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Examples

STATEMENTS	TRUTH VALUES
1. "If $1 = 1$, then $3 = 3$."	TRUE
2. "If $1 = 1$, then $2 = 3$."	FALSE
3. "If $1 = 0$, then $3 = 3$."	TRUE
4. "If $1 = 2$, then $2 = 3$."	TRUE
5. "If $1 = 1$, then $1 = 2$ and $2 = 3$."	FALSE
6. "If $1 = 3$ or $1 = 2$ then $3 = 3$."	TRUE

Alternative Ways of Expressing Implications

- “If p then q ”
- “ p implies q ”
- “if p , q ”
- “ p only if q ”
- “ p is sufficient for q ”
- “not p unless q ”
- “ q follows from p ”
- “ q if p ”
- “ q whenever p ”
- “ q is necessary for p ”

Exercise

- Your guarantee is good **only if** you bought your CD less than 90 days ago.
 If your guarantee is good, **then** you must have bought your CD player less than 90 days ago.
- To get tenure as a professor, it is **sufficient** to be world-famous.
 If you are world-famous, **then** you will get tenure as a professor.
- That you get the job implies that you have the best credentials.
 If you get the job, **then** you have the best credentials.
- It is necessary to walk 8 miles to get to the top of the peak.
 If you get to the top of the peak, **then** you must have walked 8 miles.

Translating English Sentences To Symbols

You do every exercise in this book and You get A on the final, implies, you get an A in the class.

Solution:

$$p \wedge q \rightarrow r$$

Getting an A on the final and doing every exercise in this book is sufficient for getting an A in the class.

Solution:

$$p \wedge q \rightarrow r$$

Translating English Sentences To Symbols

Let p , q , and r be the propositions:

p = “you have the flu”

q = “you miss the final exam”

r = “you pass the course”

$$p \rightarrow q$$

If you have flu, then you will miss the final exam.

$$\sim q \rightarrow r$$

If you don't miss the final exam, then you will pass the course.

$$\sim p \wedge \sim q \rightarrow r$$

If you neither have flu nor miss the final exam, then you will pass the course.

Hierarchy of Operations for Logical Connectives

1) $\sim$ (negation)

2) $\wedge$ (conjunction), $\vee$ (disjunction)

3) $\rightarrow$ (conditional)

Truth Table -- Example 1

$$p \vee \sim q \rightarrow \sim p$$

$p \vee \sim q \rightarrow \sim p$ means $(p \vee (\sim q)) \rightarrow (\sim p)$

p	q	$\sim q$	$\sim p$	$p \vee \sim q$	$p \vee \sim q \rightarrow \sim p$
T	T	F	F		
T	F	T	F		
F	T	F	T		
F	F	T	T		

Truth Table -- Example 1

$$p \vee \sim q \rightarrow \sim p$$

$p \vee \sim q \rightarrow \sim p$ means $(p \vee (\sim q)) \rightarrow (\sim p)$

p	q	$\sim q$	$\sim p$	$p \vee \sim q$	$p \vee \sim q \rightarrow \sim p$
			F	T	F
			F	T	F
			T	F	T
			T	T	T

Truth Table -- Example 2

$$(p \rightarrow q) \wedge (\sim p \rightarrow r)$$

p	q	r	$p \rightarrow q$	$\sim p$	$\sim p \rightarrow r$	$(p \rightarrow q) \wedge (\sim p \rightarrow r)$
T	T	T	T			
T	T	F	T			
T	F	T	F			
T	F	F	F			
F	T	T	T			
F	T	F	T			
F	F	T	T			
F	F	F	T			

Truth Table -- Example 2

$$(p \rightarrow q) \wedge (\sim p \rightarrow r)$$

p	q	r	$p \rightarrow q$	$\sim p$	$\sim p \rightarrow r$	$(p \rightarrow q) \wedge (\sim p \rightarrow r)$
T	T	T	T	F	T	T
T	T	F	T	F	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	F	T	T	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

Truth Table -- Example 3

$$p \rightarrow q \equiv \sim q \rightarrow \sim p$$

p	q	$\sim q$	$\sim p$	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

Implication Law -- Truth Table -- Example 4

$$p \rightarrow q \equiv \sim p \vee q$$

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Negation of a Conditional Statement

Since $p \rightarrow q \equiv \sim p \vee q$ therefore

$$\sim (p \rightarrow q) \equiv \sim (\sim p \vee q)$$

$$\equiv \sim (\sim p) \wedge (\sim q) \quad \text{De Morgan's law}$$

$$\equiv p \wedge \sim q \quad \text{Double Negative law}$$

Example 1 – Negate and find the alternative representation.

- 1) If Ali lives in Pakistan then he lives in Lahore.
- 2) If my car is in the repair shop, then I cannot get to class.
- 3) If x is prime then x is odd or x is 2.
- 4) If n is divisible by 6, then n is divisible by 2 and n is divisible by 3.

Solutions:

- 1) Ali lives in Pakistan and he does not live in Lahore.
- 2) My car is in the repair shop and I can get to class.
- 3) x is prime but x is not odd and x is not 2.
- 4) n is divisible by 6 but n is not divisible by 2 or by 3.

Inverse of a Conditional Statement

The inverse of the conditional statement $p \rightarrow q$
is $\sim p \rightarrow \sim q$

For instance; for an implication $p \rightarrow q$,

Its **inverse** is: $\neg p \rightarrow \neg q$

Inverse of a Conditional Statement

$p \rightarrow q$ is not equivalent to $\sim p \rightarrow \sim q$

p	q	$p \rightarrow q$	$\sim p$	$\sim q$	$\sim p \rightarrow \sim q$
T	T	T	F	F	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T

Writing Inverse of a Conditional Statement

1. If Today is Friday, Then $2+3 = 5$.
 - If today is not Friday, Then $2 + 3 \neq 5$
2. If it Snows today, I will ski tomorrow
 - If it does not snow today, I will not ski tomorrow.
3. If P is a square, then P is a rectangle.
 - If P is not a square, then P is not a rectangle.
4. If my car is in the repair shop, then I can get to class.
 - If my car is not in the repair shop, then I shall get to the class.

Converse of a Conditional Statement

The converse of the conditional statement $p \rightarrow q$
is $q \rightarrow p$

For instance; for an implication $p \rightarrow q$,

Its **converse** is: $q \rightarrow p$

Converse of a Conditional Statement

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Writing Converse of a Conditional Statement

1. If Today is Friday, Then $2+3 = 5$.
 - If $2 + 3 = 5$, Then today is Friday.
2. If it Snows today, I will ski tomorrow
 - I will not ski tomorrow If it does snow today.
3. If P is a square, Then P is a rectangle.
 - If P is a rectangle, Then P is a square.
4. If my car is in the repair shop, Then I can get to class.
 - If I go to class, Then my car is in the repair shop.

Contrapositive of a Conditional Statement

The contrapositive of the conditional statement
 $p \rightarrow q$ is

$$\sim q \rightarrow \sim p$$

A conditional and its contrapositive are equivalent.
Symbolically,

$$p \rightarrow q \equiv \sim q \rightarrow \sim p$$

For instance; for an implication $p \rightarrow q$,

Its **contrapositive** is: $\neg q \rightarrow \neg p$

Writing Inverse of a Conditional Statement

1. If Today is Friday, Then $2+3 = 5$.
 - If $2 + 3 \neq 5$, Then today is not Friday.
2. If it Snows today, I will ski tomorrow
 - I will not ski tomorrow only If it does not snow today.
3. If P is a square, then P is a rectangle.
 - If P is not a rectangle, then P is not a square.
4. If my car is in the repair shop, then I can get to class.
 - If I get to the class, then my car is not in the repair shop.

Converse, Inverse, Contrapositive

Some terminology, for an implication $p \rightarrow q$

- Its **converse** is: $q \rightarrow p$
- Its **inverse** is: $\neg p \rightarrow \neg q$
- Its **contrapositive** is: $\neg q \rightarrow \neg p$

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$ (converse)	$\neg p \rightarrow \neg q$ (inverse)	$\neg q \rightarrow \neg p$ (contrapositive)
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

SAME

Example of Converse, Inverse, Contrapositive

Write the **converse, inverse and contrapositive** of the statement

“if $x \neq 0$, then John is a programmer”

Example of Converse, Inverse, Contrapositive

“if $x \neq 0$, then John is a programmer”

- **Converse:** “if John is a programmer, then $x \neq 0$ ”
- **Inverse:** “if $x = 0$, then John is not a programmer”
- **Contrapositive:** “if John is not a programmer,
then $x = 0$ ”

Takeaway

1. An implication is logically equivalent to its contrapositive not inverse and converse.
2. The inverse and converse of an implication are logically equivalent.

Bi-Conditional Statement

If **p** and **q** are statement variables, the **bi-condition** of p and q is “**p if and only if q**” and is denoted as “ **$p \leftrightarrow q$** ”

The word “**if and only if**” are sometimes abbreviated “**iff**”.

The double headed arrow “ **$\leftrightarrow$** ” is bi-conditional operator.

Truth Table for Bi-Conditional Statement $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

Bi-Conditional Statement $p \leftrightarrow q$ -- Example

1. $1 + 1 = 3$ if and only if earth is flat True
2. Sky is blue if and only if $1 = 0$ False
3. Milk is white if and only if birds lay eggs True
4. 33 is divisible by 4 iff horse has four legs False
5. $x > 5$ iff $x^2 > 25$ False

Bi-Conditional Statement – Logical Equivalence

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T			
T	F	F			
F	T	F			
F	F	T			

Bi-Conditional Statement – Logical Equivalence

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Rephrasing Bi-Conditional

$p \leftrightarrow q$ is also expressed as:

“p is **necessary** and **sufficient** for q”

“If p **then** q, and **conversely**”

“p is **equivalent** to q”

Bi-Conditional – Examples

1. **If** it is hot outside you buy an ice cream cone, **and if** you buy an ice cream cone it is hot outside.
 - **Sol:** You buy an ice cream cone **if and only if** it is hot outside.
2. For you to win the contest it is **necessary and sufficient** that you have the only winning ticket.
 - **Sol:** You win the contest **if and only if** you hold the only winning ticket.
3. If you read the news paper every day, you will be informed and conversely.
 - **Sol:** You will be informed **if and only if** you read the news paper every day.

Truth Table for $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p) \equiv t$

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim q \rightarrow \sim p$	$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$
T	T					
T	F					
F	T					
F	F					

Truth Table for $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p) \equiv t$

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim p \rightarrow \sim q$	$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Truth Table for $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow q)$

p	q	r	$p \leftrightarrow q$	$r \leftrightarrow q$	$(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow q)$
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

Truth Table for $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow q)$

p	q	r	$p \leftrightarrow q$	$r \leftrightarrow q$	$(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow q)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	T	F
F	T	T	F	T	F
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	T	T

Truth Table for $p \wedge \sim r \leftrightarrow q \vee r$

p	q	r	$\sim r$	$p \wedge \sim r$	$q \vee r$	$p \wedge \sim r \leftrightarrow q \vee r$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Truth Table for $p \wedge \sim r \leftrightarrow q \vee r$

p	q	r	$\sim r$	$p \wedge \sim r$	$q \vee r$	$p \wedge \sim r \leftrightarrow q \vee r$
T	T	T	F	F	T	F
T	T	F	T	T	T	T
T	F	T	F	F	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	T	F	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	F	T

Truth Table for $\sim p \leftrightarrow q \equiv p \leftrightarrow \sim q$

p	q	$\sim p$	$\sim q$	$\sim p \leftrightarrow q$	$p \leftrightarrow \sim q$
T	T	F	F		
T	F	F	T		
F	T	T	F		
F	F	T	T		

Truth Table for $\sim p \leftrightarrow q \equiv p \leftrightarrow \sim q$

p	q	$\sim p$	$\sim q$	$\sim p \leftrightarrow q$	$p \leftrightarrow \sim q$
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	F	F

Truth Table for $\sim(p \oplus q) \equiv p \leftrightarrow q$

p	q	$p \oplus q$	$\sim(p \oplus q)$	$p \leftrightarrow q$
T	T			
T	F			
F	T			
F	F			

Truth Table for $\sim(p \oplus q) \equiv p \leftrightarrow q$

p	q	$p \oplus q$	$\sim(p \oplus q)$	$p \leftrightarrow q$
T	T	F		
T	F	T		
F	T	T		
F	F	F		

Truth Table for $\sim(p \oplus q) \equiv p \leftrightarrow q$

p	q	$p \oplus q$	$\sim(p \oplus q)$	$p \leftrightarrow q$
T	T	F	T	
T	F	T	F	
F	T	T	F	
F	F	F	T	

Truth Table for $\sim(p \oplus q) \equiv p \leftrightarrow q$

p	q	$p \oplus q$	$\sim(p \oplus q)$	$p \leftrightarrow q$
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	T	T

Laws of Logic

1. *Commutative Law:* $p \leftrightarrow q \equiv q \leftrightarrow p$

2. *Implication Law:* $p \rightarrow q \equiv \sim p \vee q$

3. *Exportation Law:* $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$

4. *Equivalence Law:* $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

5. *Reductio ad Absurdum:* $p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$

Applications

Prove that $p \wedge \sim q \rightarrow r \equiv \sim(p \wedge \sim q) \vee r$

Solution.

$$p \wedge \sim q \rightarrow r \equiv (p \wedge \sim q) \rightarrow r$$

Order of Operation

$$\equiv \sim(p \wedge \sim q) \vee r$$

Implication Law

Applications

Prove that $(p \rightarrow r) \leftrightarrow (q \rightarrow r) \equiv [\sim(\sim p \vee r) \vee (\sim q \vee r)] \wedge [\sim(\sim q \vee r) \vee (\sim p \vee r)]$

Solution.

$(p \rightarrow r) \leftrightarrow (q \rightarrow r) \equiv (\sim p \vee r) \leftrightarrow (\sim q \vee r)$ *Implication*

$\equiv [(\sim p \vee r) \rightarrow (\sim q \vee r)] \wedge [(\sim q \vee r) \rightarrow (\sim p \vee r)]$ *Equivalence of bi-conditional*

$\equiv [\sim(\sim p \vee r) \vee (\sim q \vee r)] \wedge [\sim(\sim q \vee r) \vee (\sim p \vee r)]$ *Implication Law*

Applications

Prove that $\sim(p \rightarrow q) \rightarrow p \equiv t$

Solution.

$$\sim(p \rightarrow r) \rightarrow p \equiv \sim[\sim(p \wedge \sim q)] \rightarrow p$$

$$\equiv (p \wedge \sim q) \rightarrow p$$

$$\equiv \sim(p \wedge \sim q) \vee p$$

$$\equiv (\sim p \vee q) \vee p$$

$$\equiv (q \vee \sim p) \vee p$$

$$\equiv q \vee (\sim p \vee p)$$

$$\equiv q \vee t$$

$$\equiv t$$

Implication Law

Double Negation

Implication Law

De Morgan's Law

Commutative Law of $\vee$

Associative law of $\vee$

Negation Law

Universal Bound Law